

SANKALP PINGALWAD

AI Engineer · ML Engineer · Data Scientist

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PROFESSIONAL SUMMARY

Final-year B.E. in Artificial Intelligence & Data Science (graduating 2026) with hands-on experience in RAG pipelines, multi-agent architectures, NLP, computer vision, and multimodal deep learning. Contributed as an AIML Engineer at Blackhole Inverse on a production News-AI system. Independently designed and deployed two publicly accessible AI applications — DocuMind AI and MailMind AI — with end-to-end ownership from architecture through CI/CD deployment. Proficient in Python, TensorFlow, Scikit-learn, LangChain, and the HuggingFace ecosystem.

EDUCATION

Vasantdada Patil Pratishthan's College of Engineering

Mumbai, India

B.E. in Artificial Intelligence & Data Science

2022 – 2026

• **Coursework:** Machine Learning · Deep Learning · NLP · Computer Vision · Statistical Analysis · DBMS · DSA

WORK EXPERIENCE

AI / ML Engineering Intern

Jan 2025 – Jun 2025

Blackhole Inverse

Remote

- Built a production News-AI pipeline (4-person team) with NLP-based article classification and multi-agent summarisation — processing 500+ articles per run at >88% labelling accuracy.
- Trained supervised ML models (Random Forest, XGBoost, SVM) with Grid Search hyperparameter tuning, improving classification AUC-ROC by ~12% over baseline.
- Engineered end-to-end feature pipelines (imputation, encoding, normalisation) for reproducible model training; maintained experiment traceability via Jupyter Notebooks and Git.

PROJECTS

DocuMind AI

GitHub

LangChain · HuggingFace Transformers · FAISS · FastAPI · Streamlit · Docker

- RAG-powered multi-document Q&A platform with semantic retrieval (FAISS + sentence-transformers) and source-attributed answers at sub-800ms latency over 100+ page PDFs.
- FastAPI REST backend (/upload, /query, /summarise), Dockerised full stack, deployed on HuggingFace Spaces with 94%+ semantic relevance on held-out factual queries.

MailMind AI

GitHub

HuggingFace Transformers · Gmail API · RL Reward Model · Streamlit · Plotly

- AI email intelligence platform with Gmail API (OAuth 2.0), NLP-based 3-class action classification (priority/reply/review), and a custom RL reward model for urgency scoring.
- Full CI/CD via GitHub Actions; lazy-loaded Transformer models reduced cold-start memory by ~40%; real-time Plotly sentiment dashboard — production-grade, publicly accessible.

Unified Context AI System

GitHub

Multi-Agent Architecture · HuggingFace · RL · Flask · SQLite · Streamlit

- 8-agent pipeline (ingestion, summarisation, intent classification, context-aware response, visual memory tagging, cognitive scoring, synthetic data gen, RL decision-making) across Email, WhatsApp, and Instagram via shared SQLite memory.
- Flask + Streamlit monitoring dashboards with /route and /intent microservice endpoints; end-to-end test suite covering 4 platform scenarios with graceful error handling.

Multimodal Emotion Recognition System

GitHub

TensorFlow · CNN · LSTM · OpenCV · Librosa · NLP

- Three-modal deep learning pipeline fusing CNN facial expression (OpenCV), MFCC speech features (Librosa), and LSTM textual sentiment for 6-class emotion recognition.
- Ablation study across early, late, and hybrid fusion strategies; late-fusion model outperformed best single-modality baseline — results documented via per-class confusion matrix.

TECHNICAL SKILLS

Languages: Python, SQL
ML Frameworks: TensorFlow, Keras, Scikit-learn, XGBoost
AI / ML: Machine Learning, Deep Learning, NLP, Multimodal Learning, Reinforcement Learning, Feature Engineering
LLM & AI Agents: LangChain, RAG, HuggingFace Transformers, Sentence Transformers, Multi-Agent Systems
Computer Vision & Speech: OpenCV, CNN, MFCC, Librosa, Facial Emotion Recognition, Speech Feature Extraction
Data Engineering: Pandas, NumPy, Data Preprocessing, Data Cleaning, Pipeline Design
Visualisation & BI: Matplotlib, Seaborn, Plotly, Power BI
Databases & Vectors: MySQL, SQLite, FAISS, ChromaDB
Deployment & APIs: FastAPI, Flask, Streamlit, REST APIs, Docker, GitHub Actions, CI/CD
Tools: Git, GitHub, Jupyter Notebook, VS Code

CERTIFICATIONS & ACHIEVEMENTS

Deep Learning for Developers — Infosys Springboard	Jun 2025
Data Analytics Job Simulation — Deloitte Australia (Forage)	Jun 2025
Smart India Hackathon 2025 — Internal Qualifier — Fasal Rakshak AI · VPP College of Engineering	2025